

COLUMBIA METRO, SC, USA

WMO: 723100

Lat: 33.942N

Lon: 81.118W

Elev: 225

StdP: 14.58

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13883

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.5	27.0	5.2	7.2	33.7	10.3	9.4	37.7	24.2	52.3	20.9	54.0	2.8	260	0.404

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	97.2	75.3	94.8	74.9	92.8	74.6	78.5	89.6	77.8	88.8	77.0	87.4	8.6	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.9	136.5	82.2	75.0	132.2	81.5	74.1	128.5	80.9	42.1	90.1	41.2	88.9	40.4	87.7	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	16.8	14.6	16.8	100.7	3.5	3.1	14.2	103.0	12.2	104.8	10.2	106.6	7.7	108.9
			15.1	80.4	3.5	1.3	12.7	81.3	10.6	82.1	8.7	82.8	6.2	83.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.8	46.0	49.6	56.4	64.8	72.7	79.5	82.3	81.3	75.8	65.1	54.4	48.4	(6)
	DBStd	14.88	9.89	9.11	8.89	7.45	6.27	4.50	3.53	3.86	5.48	7.73	8.26	9.16	(7)
	HDD50	544	200	111	40	2	0	0	0	0	0	2	46	143	(8)
	HDD65	2374	592	436	290	95	13	0	0	0	3	96	332	517	(9)
	CDD50	5941	75	100	239	446	705	886	1001	971	775	471	177	95	(10)
	CDD65	2297	3	5	24	89	253	437	536	506	328	100	12	4	(11)
	CDH74	23125	13	44	256	950	2490	4474	5810	5248	2896	832	91	21	(12)
	CDH80	10133	0	3	42	273	989	2127	2867	2452	1168	206	6	0	(13)
(14) Wind	WSAvg	5.9	6.3	6.5	7.0	7.2	6.4	5.8	5.5	5.0	5.3	5.0	4.9	5.4	(14)
(15) Precipitation	PrecAvg	49.8	4.6	4.1	4.9	3.2	3.6	4.8	5.6	6.1	3.7	3.0	2.8	3.5	(15)
	PrecMax	70.5	9.3	8.7	10.9	6.9	8.9	14.8	17.5	15.0	7.9	11.7	6.3	8.5	(16)
	PrecMin	34.3	0.8	0.9	0.6	0.3	0.7	0.7	0.6	1.0	0.1	0.0	0.4	0.6	(17)
	PrecStd	9.1	2.1	2.0	2.6	1.8	2.0	3.1	3.7	3.6	2.0	3.0	1.7	2.0	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.6	79.7	84.2	89.2	95.4	99.0	100.2	99.8	95.2	88.7	80.5	76.9	(19)
		MCWB	64.2	65.1	64.3	67.6	71.3	74.6	75.8	75.5	73.2	73.0	66.7	65.4	(20)
	2%	DB	70.7	74.1	80.4	86.3	91.6	95.8	97.2	96.1	92.5	85.2	76.7	72.6	(21)
		MCWB	60.9	60.9	63.2	66.3	70.7	74.4	76.1	75.7	73.0	70.0	64.0	63.6	(22)
	5%	DB	66.8	70.1	76.6	83.4	88.8	93.1	94.5	93.6	89.8	82.0	73.2	68.3	(23)
		MCWB	58.9	58.5	60.6	65.4	70.2	74.2	75.7	75.7	72.4	68.2	62.3	61.4	(24)
	10%	DB	62.7	65.9	72.6	79.9	85.8	90.5	92.2	91.0	87.0	78.7	69.7	64.5	(25)
		MCWB	56.2	56.4	58.9	63.3	69.0	73.4	75.6	75.1	71.8	67.0	61.5	58.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.9	67.7	68.2	71.7	75.9	78.5	80.3	79.8	77.7	76.6	71.1	69.1	(27)
		MCDB	71.7	73.9	77.6	82.1	84.7	90.3	92.5	91.0	86.8	82.3	75.0	73.4	(28)
	2%	WB	63.8	64.6	65.8	69.9	74.1	77.3	78.8	78.4	76.3	74.3	68.0	65.8	(29)
		MCDB	68.7	71.7	75.0	79.8	84.2	89.4	90.2	89.4	84.9	80.4	72.7	70.0	(30)
	5%	WB	60.8	61.5	64.0	68.3	72.7	76.3	77.9	77.7	75.3	72.0	65.8	62.9	(31)
		MCDB	65.3	67.5	72.8	78.0	83.0	87.7	89.2	88.3	83.5	78.3	70.7	67.4	(32)
	10%	WB	57.3	58.4	61.9	66.4	71.4	75.3	77.0	76.8	74.2	69.6	63.1	59.5	(33)
		MCDB	61.5	64.0	69.3	76.0	81.4	85.9	87.9	86.6	82.1	76.1	68.5	64.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	21.1	21.9	23.1	23.3	21.7	19.6	19.1	18.3	18.9	22.0	23.3	20.9	(35)
		MCDBR	24.8	26.5	27.2	26.6	24.6	22.9	22.0	21.5	22.8	24.0	26.4	24.1	(36)
	5% WB	MCWBR	16.5	15.7	13.0	10.7	8.6	6.8	6.2	5.8	7.2	11.0	15.4	16.1	(37)
		MCWBR	20.5	22.4	21.7	20.9	20.1	19.6	19.2	18.5	17.4	17.1	19.0	20.0	(38)
(40) Clear-Sky Solar Irradiance	taub		0.324	0.333	0.371	0.412	0.477	0.510	0.556	0.528	0.445	0.402	0.352	0.322	(40)
		taud	2.488	2.461	2.382	2.261	2.127	2.078	1.972	2.037	2.256	2.341	2.441	2.531	(41)
	Ebn at Noon		281	291	286	278	260	251	238	243	259	262	268	276	(42)
		Edh at Noon	27	31	36	43	50	52	58	53	41	35	29	25	(43)
(44) All-Sky Solar Radiation	RadAvg		880	1080	1422	1789	1949	2028	1927	1755	1512	1264	967	766	(44)
	RadStd		46	114	105	131	151	142	128	87	130	151	76	71	(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+1.21	+1.21	N/A	N/A	N/A	N/A	N/A	-173	+418	+286
Regional (60 neighbors)		+1.06	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+340	+239

Nomenclature: See separate page